NUMERICAL COMPUTATION OF HADAMARD FINITE PART OF SOME HYPERSINGULAR INTEGRALS IN 1, 2 AND 3 DIMENSIONS

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ABSTRACT

Data mining is a good way to find the relationship between raw data and predict the target we want which is also widely used in different field nowadays. In this project, we implement a lots of technology and method in data mining to predict the sale of an item based on its previous sale. We create a strong model to predict the sales. After evaluating this model, we conclude that this model can be used in normal life for future sale's prediction.

1 Introduction

Our project is a competition on Kaggle (Predict Future Sales). We are provided with daily historical sales data (including each products' sale date, block ,shop price and amount). And we will use it to forecast the total amount of each product sold next month. Because of the list of shops and products slightly changes every month. We need to create a robust model that can handle such situations.

2 Notations

For every physical quantity, the corresponding reference space quantity is denoted by a hat, so that, for example, if $\mathbf{x}$ is a point on a d-dimensional variety, then $\hat{\mathbf{x}}$ is the corresponding point in the reference space. If u is a function defined on the variety, then $\hat{u}$ is the corresponding function in the reference space. Furthermore, we denote by $⨍$ the Hadamard finite part of an integral.

3 Model presentation

Let's consider an finite part integral of the form

$$I = ⨍_{\hat{\tau}} f(\hat{\mathbf{y}}) \mathrm{d}\hat{\mathbf{y}}, \qquad f(\hat{\mathbf{y}}) \underset{\hat{y} \to \hat{x}}{\sim} \frac{1}{||\hat{\mathbf{y}} - \hat{\mathbf{x}}||^{d+1}}. \tag{1}$$

4 Injecting Richardson extrapolation into Guiggiani's final formula

4.1 Richardson extrapolation - general idea

For any angle θ, let's consider a finite sequence of n radial points $\rho_i(\theta)$ such that

$$0 < \rho_1(\theta) < \rho_2(\theta) < \cdots < \rho_n(\theta) < \hat{\rho}(\theta), \tag{2}$$

Recall: we can express the two first terms of the Laurent expansion of the complete polar kernel $F(\rho, \theta) = \rho K(\chi(\hat{\mathbf{x}}), \chi(\hat{\mathbf{y}}(\rho, \theta)))J(\hat{\mathbf{y}}(\rho, \theta))\hat{u}(\hat{\mathbf{y}}(\rho, \theta))$ as the following limits

$$F_{-2}(\theta) = \lim_{\rho \to 0} \rho^2 F(\rho, \theta), \qquad F_{-1}(\theta) = \lim_{\rho \to 0} \rho \left(F(\rho, \theta) - \frac{F_{-2}(\theta)}{\rho^2} \right). \tag{3}$$

such that we can write the complete polar kernel as

$$F(\rho, \theta) = \frac{F_{-2}(\theta)}{\rho^2} + \frac{F_{-1}(\theta)}{\rho} + \underset{\rho \to 1}{O}(1) \tag{4}$$

Let $c_{1,i}$ and $c_{2,i}$ be the coefficients used in the Richardson extrapolation of order n to compute the previous limits, so that we can write

$$\boxed{F_{-2}(\theta) \approx \widetilde{F}_{-2}(\theta) = \sum_{i=1}^{n} c_{2,i} \rho_i^2(\theta) F(\rho_i(\theta), \theta)}, \qquad F_{-1}(\theta) \approx \widetilde{F}_{-1}(\theta) = \sum_{i=1}^{n} c_{1,i} \rho_i(\theta) \left(F(\rho_i(\theta), \theta) - \frac{\widetilde{F}_{-2}(\theta)}{\rho_i^2(\theta)} \right). \tag{5}$$

By injecting the formula of $\widetilde{F}_{-2}(\theta)$ into the formula of $\widetilde{F}_{-1}(\theta)$, so that we can write $\widetilde{F}_{-1}(\theta)$ only as a linear combination of the complete polar kernel evaluated at the radial points $\rho_i(\theta)$, we obtain

$$\widetilde{F}_{-1}(\theta) = \sum_{i=1}^{n} c_{1,i} \rho_i(\theta) F(\rho_i(\theta), \theta) - \sum_{i=1}^{n} c_{1,i} \frac{1}{\rho_i(\theta)} \sum_{j=1}^{n} c_{2,j} \rho_j^2(\theta) F(\rho_j(\theta), \theta).$$

By letting $P_n(\theta) = \sum_{k=1}^{n} \frac{1}{\rho_k(\theta)}$ be a constant that is a polynomial in $\cos\theta$ and $\sin\theta$, we can write the previous formula as

$$\boxed{\widetilde{F}_{-1}(\theta) = \sum_{i=1}^{n} F(\rho_i(\theta), \theta) \rho_i(\theta) \Big(c_{1,i} - c_{2,i} P_n(\theta) \rho_i(\theta) \Big)} \tag{6}$$

4.2 Lagrange basis method

For this section and the following, we will no longer use θ in the notation, it will be implicit. Let's denote the regular function $G(\rho) = \rho^2 F(\rho)$, so that $G(0) = F_{-2}$ and $G'(0) = F_{-1}$. Let $\ell_i(\rho)$ be the Lagrange basis of degree $n - 1$ associated to the radial points ρ_i, defined as

$$\ell_i(\rho) = \prod_{j \neq i} \frac{\rho - \rho_j}{\rho_i - \rho_j}, \qquad i = 1, \ldots, n. \tag{7}$$

The lagrange interpolant of G and its derivative are given by

$$\Pi_{n-1} G(\rho) = \sum_{i=1}^{n} G(\rho_i) \ell_i(\rho), \qquad (\Pi_{n-1} G)'(\rho) = \sum_{i=1}^{n} G(\rho_i) \ell_i'(\rho). \tag{8}$$

So that the numerical Laurent's coefficients, denoted $\widetilde{F}_{-2}$ and $\widetilde{F}_{-1}$, are simply the evaluation of the interpolant and its derivative at the origin $\rho = 0$:

$$\widetilde{F}_{-2} = \Pi_{n-1}G(0) = \sum_{i=1}^{n} G(\rho_i)\ell_i(0), \qquad \widetilde{F}_{-1} = (\Pi_{n-1}G)'(0) = \sum_{i=1}^{n} G(\rho_i)\ell_i'(0). \tag{9}$$

with $\ell_i(0) = \prod_{j\neq i} \frac{\rho_j}{\rho_j - \rho_i}$ and $\ell_i'(0) = \ell_i(0)\left(\frac{1}{\rho_i} - P_n\right)$, so that we can finally write the Laurent's coefficients as

$$\widetilde{F}_{-2} = \sum_{i=1}^{n} \ell_i(0)\rho_i^2 F(\rho_i), \qquad \widetilde{F}_{-1} = \sum_{i=1}^{n} \ell_i(0)\rho_i\left(1 - P_n\rho_i\right)F(\rho_i). \tag{10}$$

This is a particular case of the Richardson extrapolation, with the two families of coefficients $c_{1,i}$ and $c_{2,i}$ collapsing into a single family of coefficients $\ell_i(0)$:

$$c_{1,i} = c_{2,i} = \ell_i(0) = c_i = \prod_{j\neq i} \frac{\rho_j}{\rho_j - \rho_i}.$$

Finally, the numerical Laurent's coefficients can be written as

$$\widetilde{F}_{-2}(\theta) = \sum_{i=1}^{n} A_i(\theta)F(\rho_i(\theta),\theta), \qquad \widetilde{F}_{-1}(\theta) = \sum_{i=1}^{n} B_i(\theta)F(\rho_i(\theta),\theta). \tag{11}$$

With $A_i(\theta) = c_i(\theta)\rho_i^2(\theta)$ and $B_i(\theta) = c_i(\theta)\rho_i(\theta)\left(1 - P_n(\theta)\rho_i(\theta)\right)$.

Each of these formula simply consist in the numerical evaluation of the polar kernel at the radial points $\rho_i(\theta)$, multiplied by some weights.

4.2.1 Properties of $A_i(\theta)$ and $B_i(\theta)$

$$\sum_{i=1}^{n} A_i = \begin{cases} \rho_1, & n = 1 \\ -\rho_1\rho_2, & n = 2 \\ 0, & n \geq 3 \end{cases}, \qquad \sum_{i=1}^{n} B_i = \begin{cases} 0, & n = 1 \\ \rho_1 + \rho_2, & n = 2 \\ 0, & n \geq 3 \end{cases} \tag{12}$$

These three cases illustrates the following definition :

$$\sum_{i=1}^{n} A_i = (\Pi_{n-1}[\rho^2])(0), \qquad \sum_{i=1}^{n} B_i = (\Pi_{n-1}[\rho^2])'(0) \tag{13}$$

4.3 Final quadrature rule

Let's then denote $I(\theta)$ the integral of the complete polar kernel over the radial direction, defined as:

$$I(\theta) = \int_0^{\hat{\rho}(\theta)} \left\{ F(\rho,\theta) - \frac{F_{-1}(\theta)}{\rho} - \frac{F_{-2}(\theta)}{\rho^2} \right\} \mathrm{d}\rho + F_{-1}(\theta)\ln\left|\frac{\hat{\rho}(\theta)}{\beta(\theta)}\right| - F_{-2}(\theta)\left\{\frac{\gamma(\theta)}{\beta^2(\theta)} + \frac{1}{\hat{\rho}(\theta)}\right\} \tag{14}$$

A final quadrature rule for it is the following:

$$I(\theta) \approx \widetilde{I}(\theta) = \sum_{i=1}^{n} \widetilde{w}_i(\theta)F(\rho_i(\theta),\theta) \tag{15}$$

With the new weights $\widetilde{w}_i(\theta)$ defined as

$$\widetilde{w}_i(\theta) = w_i(\theta) - A_i(\theta)\left\{W_{-2,n}(\theta) + \frac{\gamma(\theta)}{\beta^2(\theta)} + \frac{1}{\hat{\rho}(\theta)}\right\} + B_i(\theta)\left\{\ln\left|\frac{\hat{\rho}(\theta)}{\beta(\theta)}\right| - W_{-1,n}(\theta)\right\}$$

and the auxiliary constants :

- $w_i(\theta) = \frac{\hat{\rho}(\theta)}{2} w_i^{\text{Ref}}$ are the weights of the reference quadrature over the segment $[-1, 1]$.
- $W_{-1,n}(\theta) = \sum_{i=1}^{n} \frac{w_i(\theta)}{\rho_i(\theta)} = \sum_{i=1}^{n} \frac{w_i^{\text{Ref}}}{1+x_i^{\text{Ref}}}$.
- $W_{-2,n}(\theta) = \sum_{i=1}^{n} \frac{w_i(\theta)}{\rho_i^2(\theta)} = \frac{2}{\hat{\rho}(\theta)} \sum_{i=1}^{n} \frac{w_i^{\text{Ref}}}{\left(1+x_i^{\text{Ref}}\right)^2}$.

Remark: in the case of Gauss-Legendre quadrature, the constants $W_{-1,n}$ and $W_{-2,n}$ can be computed analytically as:

$$W_{-1,n} = 2H_n, \qquad W_{-2,n}(\theta) = \frac{2}{\hat{\rho}(\theta)} n(n+1), \qquad P_n(\theta) = \frac{n(n+1)}{\hat{\rho}(\theta)} \tag{16}$$

With $H_n = \sum_{i=1}^{n} \frac{1}{i}$ the n-th harmonic number. These last two formulas are, for the moment, a conjecture. (Elaborate to prove it?)

4.4 Properties of the new weights $\widetilde{w}_i(\theta)$

$$\text{For } n \geq 3, \quad \sum_{i=1}^{n} \widetilde{w}_i(\theta) = \hat{\rho}(\theta) \tag{17}$$

This is easy to understand: either look at the definition of $\widetilde{w}_i(\theta)$ and use the properties of $A_i(\theta)$ and $B_i(\theta)$, or integrate the constant function $F(\rho, \theta) = 1$ between 0 and $\hat{\rho}(\theta)$, which is equivalent. In a more general way, if the integrand F is not singular (i.e. $F \sim \mathcal{O}(1)$), then the quadrature rule is the standard quadrature as if reference quadrature was used to integrate the function F between 0 and $\hat{\rho}(\theta)$. If the integrand is singular as $1/\rho$, the customed quadrature just performs a Cauchy principal value. The true ingredients are the values of the Laurent coefficients F_{-1} and F_{-2}, which are computed numerically; they can be zero (up to the algorithm precision) or not, depending on the considered singularity.

4.5 Convergence and error study

4.5.1 General idea

To begin our reasoning, assume that we have computed the Laurent's coefficients up to a relative error ϵ_{-k}, $k \in \{1, 2\}$:

$$\widetilde{F}_{-k} = F_{-k}(1 + \epsilon_{-k}), \qquad k \in \{1, 2\} \tag{18}$$

Now let's denote

$$R(\rho, \theta) = F(\rho, \theta) - \frac{F_{-1}(\theta)}{\rho} - \frac{F_{-2}(\theta)}{\rho^2} \tag{19}$$

$$\widetilde{R}(\rho, \theta) = F(\rho, \theta) - \frac{\widetilde{F}_{-1}(\theta)}{\rho} - \frac{\widetilde{F}_{-2}(\theta)}{\rho^2} \tag{20}$$

$$\tag{21}$$

$$I = \int_0^{2\pi} \int_0^{\hat{\rho}(\theta)} R(\rho, \theta) \mathrm{d}\rho \mathrm{d}\theta, \qquad \widetilde{I} = \int_0^{2\pi} \int_0^{\hat{\rho}(\theta)} \widetilde{R}(\rho, \theta) \mathrm{d}\rho \mathrm{d}\theta \tag{22}$$

On one hand, after the theory of the Bernstein's ellipses, if we assume that R and $\widetilde{R}$ are analytic, we can make a error majoration regarding the Gauss-Legendre quadrature of I and $\widetilde{I}$:

$$|I - Q_n[R]| \leq \frac{64}{15} \frac{M}{(1-\rho_B^{-2})\rho_B^{2n}} \tag{23}$$

$$\left|\widetilde{I} - Q_n[\widetilde{R}]\right| \leq \frac{64}{15} \frac{\widetilde{M}}{(1-\widetilde{\rho}_B^{-2})\widetilde{\rho}_B^{2n}} \tag{24}$$

On the other hand, we can substract $\widetilde{I}$ to I to get:

$$|I - \widetilde{I}| \leq C(\hat{x}) \sum_{k=1}^{2} \frac{\epsilon_{-k}}{\rho_{\min}^k} ||F_{-k}||_\infty \tag{25}$$

where $C(\hat{x}) = \int_0^{2\pi} \hat{\rho}(\theta)\mathrm{d}\theta$.

Thus,

$$|I - Q_n[\widetilde{R}]| = |I - \widetilde{I} + \widetilde{I} - Q_n[\widetilde{R}]| \quad \leq |I - \widetilde{I}| + |\widetilde{I} - Q_n[\widetilde{R}]| \leq C(\hat{x}) \sum_{k=1}^{2} \frac{\epsilon_{-k}}{\rho_{\min}^k} ||F_{-k}||_\infty + \frac{64}{15} \frac{\widetilde{M}}{(1-\widetilde{\rho}_B^{-2})\widetilde{\rho}_B^{2n}} \tag{26}$$

4.5.2 Richardson extrapolation

We can show that the following majorations hold for any sequence $\rho_i(\theta)$ of n radial points:

$$|\widetilde{F}_{-2}(\theta) - F_{-2}(\theta)| \leq \frac{||R^{(n)}||_\infty(\theta)}{n!} \prod_{i=1}^{n} \rho_i(\theta), \qquad |\widetilde{F}_{-1}(\theta) - F_{-1}(\theta)| \leq P_n(\theta) \frac{||R^{(n+1)}||_\infty(\theta)}{n!} \prod_{i=1}^{n} \rho_i(\theta) \tag{27}$$

Which shows that the speed of convergence is conditionned by the product of the radial points $\prod_{i=1}^{n} \rho_i(\theta)$. In the case of a geometric sequence $\rho_i(\theta) = \hat{\rho}(\theta) q^{i-1}$, we have $\prod_{i=1}^{n} \rho_i(\theta) = \hat{\rho}^n(\theta) q^{\frac{n(n-1)}{2}}$, which is a very fast convergence. In the case of a sequence of Gauss-Legendre quadrature points, we have $\prod_{i=1}^{n} \rho_i(\theta) = \hat{\rho}^n(\theta) \frac{(n!)^2}{(2n)!}$, which is a slower convergence (see Figure 1).

That's why geometric sequences are still preferred over Gauss-Legendre quadrature points for pure Richardson extrapolation.

4.5.3 Gauss-Legendre quadrature

To delve deeper into these general ideas, let's consider :

$$\widetilde{I}_{\epsilon,h,n}(\theta) = Q_n[\widetilde{R}_{\epsilon,h}](\theta) = \sum_{i=1}^{n} w_i(\theta) \left\{ F_h(\rho_i(\theta), \theta) - \frac{\widetilde{F}_{-1,\epsilon,h}(\theta)}{\rho_i(\theta)} - \frac{\widetilde{F}_{-2,\epsilon,h}(\theta)}{\rho_i^2(\theta)} \right\} \tag{28}$$

$$= \sum_{i=1}^{n} \left\{ F_h(\rho_i(\theta), \theta) - \frac{F_{-1,h}(\theta)}{\rho_i(\theta)} - \frac{F_{-2,h}(\theta)}{\rho_i^2(\theta)} \right\} w_i(\theta) - \sum_{k=1}^{2} F_{-k,h}(\theta) \epsilon_{-k} \sum_{i=1}^{n} \frac{w_i(\theta)}{\rho_i^k(\theta)} \tag{29}$$

$$= Q_n[R_h](\theta) - \sum_{k=1}^{2} F_{-k,h}(\theta) \epsilon_{-k} W_{-k,n}(\theta) \tag{30}$$

Where h is the mesh size. Thanks to standard results on the Gauss-Legendre quadrature for any function in $\mathcal{C}^{2n}([0, \hat{\rho}(\theta)])$, we have the following error majoration (deleting the θ dependency for clarity):

$$Q_n[R_h] = \int_0^{\hat{\rho}} R_h(\rho)\mathrm{d}\rho + \frac{2^{2n+1}(n!)^4}{(2n+1)[(2n)!]^3} R_h^{(2n)}(\rho_0) \tag{31}$$

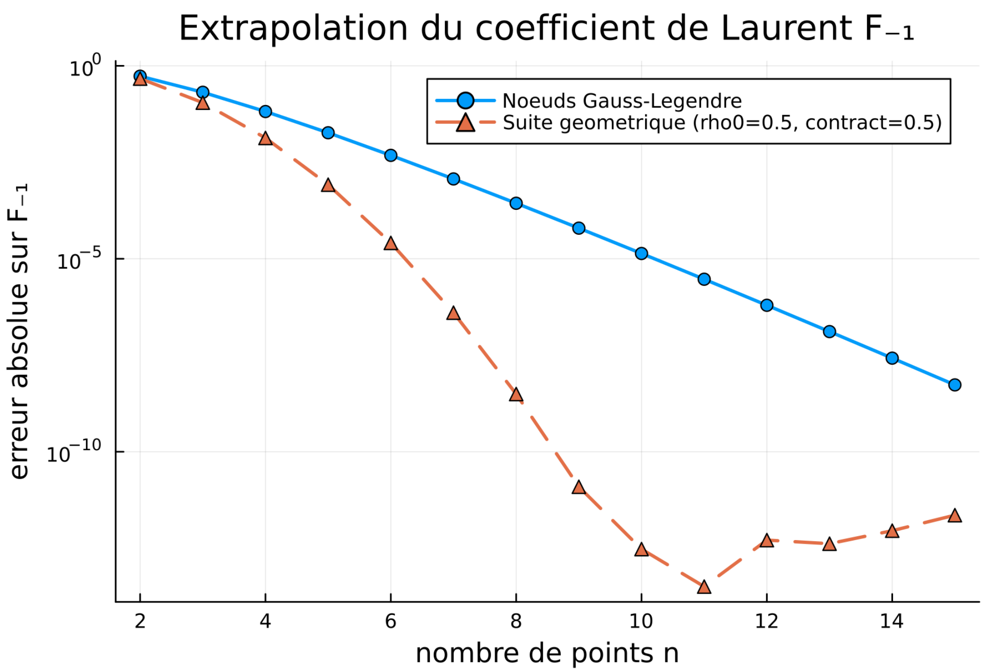

Figure 1: Convergence comparison between geometric and Gauss-Legendre sequences for the function $\rho \to 1/(1+\rho)$

for some $\rho_0 \in [0, \hat{\rho}]$. Our next goal is to show that the error term $R_h^{(2n)}(\rho_0)$ converges with the mesh size h and the number of quadrature points n. By using $G_h(\rho) = \rho^2 F_h(\rho) = \mathcal{O}_{\rho\to 0}(1)$, we can straightforwardly express R_h in terms of G, and bound its derivatives, assuming, as Guiggiani, that F_h is analytic in a neighborhood of the origin:

$$R_h(\rho) = \sum_{k=0}^{+\infty} F_k \rho^k, \qquad \Longrightarrow \qquad R_h^{(2n)}(\rho) = \sum_{k=2n}^{+\infty} F_k \frac{k!}{(k-2n)!} \rho^{k-2n} \tag{32}$$

$$\tag{33}$$

with $F_k = \dfrac{G_h^{(k+2)}(0)}{(k+2)!} \leq \dfrac{1}{(k+2)!} ||G_h^{(k+2)}||_{\infty,[0,\hat{\rho}]}$ and $\rho^{k-2n} \leq \hat{\rho}^{k-2n}$, so that we can write

$$|R_h^{(2n)}(\rho)| \leq \sum_{k=2n}^{+\infty} k! \, \frac{||G_h^{(k+2)}||_{\infty,[0,\hat{\rho}]}}{(k+2)! \, (k-2n)!} \hat{\rho}^{k-2n} \tag{34}$$

BOUNDING OF G AND ITS DERIVATIVES: let us defined the following functions:

$$\phi_h \; : \; \begin{array}{ccc} \hat{\tau} \subset \mathbb{R}^{d-1} & \to & \mathbb{R} \\ \hat{x} & \mapsto & \det M_h(\hat{x}) \end{array}, \qquad \psi_{\hat{x}} \; : \; \begin{array}{ccc} \mathbb{R} \times [0, 2\pi[& \to & \mathbb{R}^{d-1} \\ (\rho, \theta) & \mapsto & \hat{x} + \rho \mathbf{u}(\theta) \end{array}$$

$$\chi_h \; : \; \begin{array}{ccc} \hat{\tau} \subset \mathbb{R}^{d-1} & \to & \tau \subset \mathbb{R}^d \\ \hat{x} & \mapsto & x \end{array}$$

where χ is the mapping from the reference space to the physical element, $\psi_{\hat{x}}$ is the mapping from polar coordinates to the reference space, and M_h is the metric tensor of the mapping χ_h, defined as:

$$M_h(\hat{x}) = D\chi_h(\hat{x})^T D\chi_h(\hat{x}), \qquad (M_h)_{ab} = \sum_{k=1}^{d-1} \frac{\partial \chi_{h,k}}{\partial \hat{x}_a} \frac{\partial \chi_{h,k}}{\partial \hat{x}_b}, \quad (a,b) \in \{1, \dots, d-1\}^2 = \{1,2\}^2 \tag{35}$$

Lemma 1: using the Faà di Bruno formula for functions of the form $h(x) = f[g_1(x), \cdots, g_m(x)]$ (see Constantine, 1996), one can write, for any function $f : \mathbb{R}^{d-1} \to \mathbb{R}$,

$$\frac{\partial^k (f \circ \psi_{\hat{x}})}{\partial \rho^k}(\rho, \theta) = \sum_{i=0}^{k} \binom{k}{i} \cos^i(\theta) \sin^{k-i}(\theta) \frac{\partial^k f}{\partial \hat{x}_1^i \partial \hat{x}_2^{k-i}}(\psi_{\hat{x}}(\rho, \theta)) \tag{36}$$

Finally, the vector $\mathbf{u}(\theta)$ is the unit trigonometric vector defined as $\mathbf{u}(\theta) = (\cos\theta, \sin\theta)$ for our case $d = 3$. We can then expand G as

$$G(\rho, \theta) = H_h(\rho, \theta) J_h \circ \psi_{\hat{x}}(\rho, \theta) \tag{37}$$

where

$$H_h(\rho, \theta) = \rho^3 K\Big(x = \chi_h(\hat{x}), y = \chi_h \circ \psi_{\hat{x}}(\rho, \theta)\Big) \hat{u} \circ \psi_{\hat{x}}(\rho, \theta), \quad J_h(\hat{x}) = \sqrt{\phi_h(\hat{x})} = \sqrt{\det M_h(\hat{x})} \tag{38}$$

As the kernel K is regularized by the term ρ^3 in H_h, we can ensure that $H_h = \mathcal{O}_{\rho\to 0}(1)$. Applying the Leipniz formula, we have

$$\frac{\partial^m G(\rho, \theta)}{\partial \rho^m} = \sum_{k=0}^{m} \binom{m}{k} \frac{\partial^k H_h(\rho, \theta)}{\partial \rho^k} \frac{\partial^{m-k} (J_h \circ \psi_{\hat{x}})}{\partial \rho^{m-k}}(\rho, \theta) \tag{39}$$

We then let

$$\begin{cases} \gamma_{h,m} = \sup\limits_{k\le m} \sup\limits_{\theta\in[0,2\pi[} \sup\limits_{\rho\in[0,\hat{\rho}]} \left\| \dfrac{\partial^k H_h(\rho, \theta)}{\partial \rho^k} \right\| = \underset{h\to 0}{O}(1) \\ M_m = \max\limits_{k\le m} \dbinom{m}{k} = \dbinom{m}{\lfloor m/2 \rfloor} \end{cases} \tag{40}$$

So that we can write

$$\left| \frac{\partial^m G(\rho, \theta)}{\partial \rho^m} \right| \le (m+1) M_m \gamma_{h,m} \sum_{k=0}^{m} \left| \frac{\partial^k (J_h \circ \psi_{\hat{x}})}{\partial \rho^k}(\rho, \theta) \right| \tag{41}$$

BOUNDING OF THE k-TH RADIAL DERIVATIVE OF $J_h \circ \psi_{\hat{x}}$: using the lemma 36, we can write

$$\frac{\partial^k (J_h \circ \psi_{\hat{x}})}{\partial \rho^k}(\rho, \theta) = \sum_{i=0}^{k} \binom{k}{i} \cos^i(\theta) \sin^{k-i}(\theta) \frac{\partial^k J_h}{\partial \hat{x}_1^i \partial \hat{x}_2^{k-i}}(\psi_{\hat{x}}(\rho, \theta)) \tag{42}$$

$$\le (k+1) M_k \sum_{i=0}^{k} \left| \frac{\partial^k J_h}{\partial \hat{x}_1^i \partial \hat{x}_2^{k-i}}(\psi_{\hat{x}}(\rho, \theta)) \right| \tag{43}$$

BOUNDING OF THE CARTHESIAN DERIVATIVES OF J_h: there's no need to expand an explicit formula for the derivatives of J_h in terms parametrization χ_h, because we can use a useful hypothesis on the mesh quality. Indeed, to ensure non-degeneracy of the elements, we can assume that

$$\exists C_Q > 0, \quad \forall h > 0, \quad \forall m > 0, \quad \forall \hat{x} \in \hat{\tau}, \quad \|D^m \chi_h(\hat{x})\| \le C_Q h^m \tag{44}$$

Where $D^m \chi(\hat{x})$ is the tensor of order m defined containing all the m-th order derivatives of χ_h at $\hat{x}$, defined as

$$(D^m \chi_h(\hat{x}))_{i,\alpha_1,\cdots\alpha_m} = \frac{\partial^m \chi_{h,i}}{\partial \hat{x}_{\alpha_1} \cdots \partial \hat{x}_{\alpha_m}}(\hat{x}), \qquad i \in \{1, 2, 3\}, \quad (\alpha_1, \cdots, \alpha_m) \in \{1, 2\}^m \tag{45}$$

In particular, for $\alpha = 1$, we have

$$\|M_h(\hat{x})\| = \left\|D\chi_h(\hat{x})^T D\chi_h(\hat{x})\right\| \leq C_Q^2 h^2. \tag{46}$$

Furthermore, the Hadamard's inequality (HI) states that

$$|\phi_h(\hat{x})| = |\det(M_h(\hat{x}))| \overset{\text{(HI)}}{\leq} \prod_{i=1}^{d-1} \|M_h(\hat{x})_{i,:}\| \leq \|M_h(\hat{x})\|^{d-1} \leq C_Q^{2(d-1)} h^{2(d-1)} \tag{47}$$

Remarks : theses inequalities holds perfectly if we consider the ∞-norm, but all the norms are equivalent in finite dimension, so we can use any norm; that leads to different majoration constants but the idea remains the same. This ensure that we can define the following function:

$$\begin{array}{rccl} \mu & : & \hat{\tau} \subset \mathbb{R}^{d-1} & \to & \mathbb{R} \\ & & \hat{x} & \mapsto & \sqrt{h^{-2(d-1)} \det M_h(\hat{x})} = h^{-(d-1)} J_h(\hat{x}) \leq C_Q^{d-1} \end{array}$$

By the non-degeneracy hypothesis, we have $\mu = \mathcal{O}_{h\to 0}(1)$ and all its derivatives are also bounded. So that we can write $J_h(\hat{x}) = h^{d-1}\mu(\hat{x})$, and

$$\left|\frac{\partial^m G(\rho,\theta)}{\partial \rho^m}\right| \leq h^{d-1}(m+1) M_m \gamma_{h,m} \sum_{k=0}^{m} (k+1) M_k \sum_{i=0}^{k} \left\|\frac{\partial^i \mu}{\partial \hat{x}^i \partial \hat{x}^{k-i}} \left(\psi_{\hat{x}}(\rho,\theta)\right)\right\| \tag{48}$$

Naturally, let now $\mu_k = \sup_{i\leq k} \left\|\frac{\partial^k \mu}{\partial \hat{x}_1^i \partial \hat{x}_2^{k-i}}\right\|_{\infty,\hat{\tau}}$ and $\lambda_m = \max_{k\leq m} \mu_k$, we have the simplified following majoration:

$$\left|\frac{\partial^m G(\rho,\theta)}{\partial \rho^m}\right| \leq h^{d-1}(m+1)^4 M_m^2 \gamma_{h,m} \lambda_m \tag{49}$$

Thus, the remainder term of Laurent's serie expansion can be bounded as

$$|R_h^{(2n)}(\rho)| \leq h^{d-1} \sum_{k=2n}^{+\infty} k! \, \frac{(k+3)^4 M_{k+2}^2 \gamma_{h,k+2} \lambda_{k+2}}{(k+2)!\,(k-2n)!} \hat{\rho}^{k-2n} \tag{50}$$

EXPLICIT MAJORATION OF $\gamma_{h,k}$: let's recall the definition of $\gamma_{h,k}$:

$$\gamma_{h,k} = \sup_{m\leq k} \sup_{\theta\in[0,2\pi[} \sup_{\rho\in[0,\hat{\rho}]} \left\|\frac{\partial^m H_h(\rho,\theta)}{\partial \rho^m}\right\| \tag{51}$$

where

$$\frac{\partial^m H_h(\rho,\theta)}{\partial \rho^m} = \frac{\partial^m}{\partial \rho^m} \left\{\rho^s K(\chi_h(\hat{x}), \chi_h \circ \psi_{\hat{x}}(\rho,\theta)) \hat{u} \circ \psi_{\hat{x}}(\rho,\theta)\right\} \tag{52}$$

$$= \sum_{k=0}^{m} \binom{m}{k} \frac{\partial^k}{\partial \rho^k} \Big(\rho^s K(\chi_h(\hat{x}), \chi_h \circ \psi_{\hat{x}}(\rho,\theta))\Big) \frac{\partial^{m-k}}{\partial \rho^{m-k}} \Big(\hat{u} \circ \psi_{\hat{x}}(\rho,\theta)\Big) \tag{53}$$

$$\overset{36}{\leq} (m+1)^2 \delta_m M_m^2 \sum_{k=0}^{m} \frac{\partial^k}{\partial \rho^k} \Big(\rho^s K(\chi_h(\hat{x}), \chi_h \circ \psi_{\hat{x}}(\rho,\theta))\Big) \tag{54}$$

where $\delta_m = \sup_{i\leq m} \left\|\frac{\partial^m \hat{u}\,(\hat{x})}{\partial \hat{x}_1^i \partial \hat{x}_2^{m-i}}\right\|_{\infty,\hat{\tau}}$ is a constant that depends only on the geometry of the element, and s is the order of singularity of the function $\rho \to K(\chi_h(\hat{x}), \chi_h \circ \psi_{\hat{x}}(\rho,\theta))$ at the origin, such that $K(\chi_h(\hat{x}), \chi_h \circ \psi_{\hat{x}}(\rho,\theta)) = \mathcal{O}_{\rho\to 0}(\rho^{-s})$. The idea was originally used for hypersingular 3D kernels, where $s = 3$ (see Guiggiani, 1992), but we write the computations with general order s because the idea remains the same. Then, we recall the shape-regularity condition which the upper bound equivalent of the non-degeneracy condition 61:

$$\exists\, c_Q > 0, \quad \forall h > 0, \quad \forall (\hat{x}, \hat{y}) \in \hat{\tau}^2, \quad \|\chi_h(\hat{x}) - \chi_h(\hat{y})\| \geq c_Q\, h\, \|\hat{x} - \hat{y}\|. \tag{55}$$

Lemma 2: We have the following useful factorization:

$$\boldsymbol{r} = \rho \boldsymbol{A}_{\hat{x}}(\rho, \theta), \quad \boldsymbol{A}_{\hat{x}}(\rho, \theta) = \sum_{m \geq 1} \frac{\rho^{m-1}}{m!} D^m \chi_h(\hat{x}) :_m \left(\boldsymbol{u}(\theta)^{m\otimes}\right) \tag{56}$$

Condition 61 can be rewritten as $\boxed{r \geq c_Q h \rho}$, which represent the bi-Lipschitz property of the parametrization χ_h.

Homogeneous kernels. When the kernel is homogeneous, i.e

$$\exists \hat{K}, K(x, y) = \frac{1}{r^s} \hat{K}(\hat{\mathbf{r}}, n_x, n_y), \quad \forall n_x, n_y \in \mathbb{S}^d, z \to \hat{K}(z, n_x, n_y) \text{ is of class } \mathcal{C}^\infty(\mathbb{R}^d), \tag{57}$$

$$\text{i.e } \forall \alpha \in \mathbb{N}^d, \quad \exists C_K(\alpha) > 0, \quad \left| \frac{\partial^{|\alpha|} \hat{K}}{\partial z^\alpha}(z, n_x, n_y) \right| \leq C_K(\alpha), \quad \forall z \in \mathbb{R}^d \tag{58}$$

Such that

$$\rho^s K(\chi_h(\hat{x}), \chi_h \circ \psi_{\hat{x}}(\rho, \theta)) = \left(\frac{\rho}{r}\right)^s \hat{K}\left(\hat{\mathbf{A}}_{\hat{x}}(\rho, \theta), n \circ \chi_h(\hat{x}), n \circ \chi_h \circ \psi_{\hat{x}}(\rho, \theta)\right) \tag{59}$$

$$= \frac{1}{\|\boldsymbol{A}_{\hat{x}}(\rho, \theta)\|^s} \hat{K}\left(\hat{\mathbf{A}}_{\hat{x}}(\rho, \theta), n \circ \chi_h(\hat{x}), n \circ \chi_h \circ \psi_{\hat{x}}(\rho, \theta)\right) \tag{60}$$

4.5.4 Completing the majoration of $\gamma_{h,k}$ - to delete

The derivation below completes the majoration of $\gamma_{h,k}$ left open above; it was worked out by Claude (Anthropic), continuing the formalism and notations introduced in this note.

A complementary non-degeneracy hypothesis. Hypothesis (44) only controls χ_h *from above*: it prevents an element from varying faster than a curve of typical scale h, but it does not prevent χ_h from being locally degenerate, i.e. from mapping two distinct points of $\hat{\tau}$ onto (near-)coincident points of τ. Controlling $\|x - y\|$ *from below* is exactly what is needed to use (**??**), whose right-hand side blows up as $y \to x$. We therefore introduce the complementary, and equally standard, non-degeneracy hypothesis:

$$\exists\, c_Q > 0, \quad \forall h > 0, \quad \forall (\hat{x}, \hat{y}) \in \hat{\tau}^2, \quad \|\chi_h(\hat{x}) - \chi_h(\hat{y})\| \geq c_Q\, h\, \|\hat{x} - \hat{y}\|. \tag{61}$$

Together, (44) (for $m = 1$, via the mean value theorem) and (61) state that χ_h is bi-Lipschitz on $\hat{\tau}$ with constants of order exactly h in both directions: this is the usual "no degenerate/flattened element" assumption of a shape-regular mesh, and it is genuinely stronger than (44) alone. Applied to $\hat{y} = \psi_{\hat{x}}(\rho, \theta) = \hat{x} + \rho \mathbf{u}(\theta)$, and using $\|\hat{x} - \hat{y}\| = \rho \|\mathbf{u}(\theta)\| = \rho$, this gives the key estimate used throughout what follows:

$$\|x - y\| = \|\chi_h(\hat{x}) - \chi_h \circ \psi_{\hat{x}}(\rho, \theta)\| \geq c_Q\, h\, \rho, \qquad \forall \rho \in [0, \hat{\rho}],\ \forall \theta \in [0, 2\pi[. \tag{62}$$

An exact cancellation at order $k = 0$. Combining (**??**) with $\alpha = 0$ and (62),

$$|K(\chi_h(\hat{x}), \chi_h \circ \psi_{\hat{x}}(\rho, \theta))| \leq C_K(0)\, \|x - y\|^{-s} \leq C_K(0)\, c_Q^{-s}\, h^{-s}\, \rho^{-s}. \tag{63}$$

Multiplying by $\rho^s \|\hat{u} \circ \psi_{\hat{x}}(\rho, \theta)\| \leq \rho^s \delta_0$ (recall $\delta_0 = \|\hat{u}\|_{\infty, \hat{\tau}}$), the factor ρ^{-s} is cancelled *exactly* by the regularizing prefactor ρ^s carried by H_h. We obtain, for every $\rho \in [0, \hat{\rho}]$ and $\theta \in [0, 2\pi[$ (the bound extends to $\rho = 0$ by continuity, consistently with $H_h = \mathcal{O}_{\rho \to 0}(1)$):

$$|H_h(\rho,\theta)| \le \Gamma_0\, h^{-s}, \qquad \Gamma_0 := C_K(0)\, c_Q^{-s}\, \delta_0. \tag{64}$$

In particular $\gamma_{h,0} \le \Gamma_0\, h^{-s}$: the correct order is h^{-s}, not $\mathcal{O}_{h\to 0}(1)$ as anticipated in (51). This is in fact expected: H_h carries the full strength of an order-s hypersingularity, now concentrated on a patch of physical size h shrinking around x, and there is no reason for this to stay bounded as $h \to 0$.

Why the same computation fails for $k \ge 1$. One could hope to bound $\gamma_{h,k}$, $k \ge 1$, by repeating the argument above on each term of $\partial_\rho^k H_h$ after expanding through Lemma 2 (**??**). This fails termwise. Differentiating $\rho \mapsto K(\chi_h(\hat{x}), \chi_h \circ \psi_{\hat{x}}(\rho,\theta))$ j times, via Lemma 36 and the (multivariate) Faà di Bruno formula, produces a sum over partitions of j of terms of the form

$$D_y^m K(x,y) \prod_{l=1}^{m} D^{k_l} \chi_h(\hat{y}), \qquad \sum_l k_l = j, \quad 1 \le m \le j,$$

i.e. a K-derivative of order m (the number of parts of the partition) times m derivatives of χ_h whose orders sum to j. Bounding each geometric factor with (44) ($\prod_l \|D^{k_l}\chi_h\| \le C_Q^m h^j$, since $\sum_l k_l = j$) and the kernel factor with (**??**)–(62) ($\|D_y^m K(x,y)\| \le C_K(m)(c_Q h\rho)^{-(m+s)}$) shows every such term is $\mathcal{O}(h^{j-m-s}\rho^{-(m+s)})$. Multiplying by the prefactor ρ^{s-k+j} that Lemma 2 attaches to the corresponding j-th order term, the net power of ρ is ρ^{j-k-m} – *strictly negative* for every admissible triple (k,j,m) with $k \ge 1$, since $j \le k$ and $m \ge 1$. Every term of the expansion, bounded individually through the triangle inequality, therefore diverges as $\rho \to 0^+$.

This does not contradict the regularity of H_h at the origin: it only shows that the terms of the Leibniz/Faà di Bruno expansion of $\partial_\rho^k(\rho^s K)$ cancel each other exactly as $\rho \to 0$, and that this cancellation – visible in the exact identity $\rho^s K = H_h/\hat{u} \circ \psi_{\hat{x}}$, an $\mathcal{O}(1)$-in-ρ quantity times h^{-s} – is destroyed as soon as K and χ_h are bounded separately and recombined with the triangle inequality. A genuinely different route is therefore needed for $k \ge 1$: differentiating $\rho^s K$ termwise loses exactly the cancellation that made (64) work.

An exact algebraic factorization, valid at every order. The $k = 0$ computation above cancelled exactly because ρ^s was cancelled *before* bounding K, not after. The fix for $k \ge 1$ is to keep it that way at every order: rather than bound K pointwise and reintroduce ρ^s afterwards through Lemma 2, we factor ρ^s out of K *algebraically*, once and for all, using the specific (pseudo-homogeneous) structure that hypersingular BEM kernels possess and from which (**??**) itself is classically derived (Guiggiani, 1992):

$$K(x,y) = \frac{\kappa(x,\omega)}{\|x-y\|^s}, \qquad \omega = \frac{y-x}{\|y-x\|} \in \mathbb{S}^{d-1}, \tag{65}$$

where, for every x, $\omega \mapsto \kappa(x,\omega)$ is of class $\mathcal{C}^{2n}(\mathbb{S}^{d-1})$ with

$$\exists\, C_\kappa(\beta) > 0, \quad \left\|D_\omega^\beta \kappa(x,\omega)\right\| \le C_\kappa(\beta), \qquad \forall\, \omega \in \mathbb{S}^{d-1},\ \beta \in \mathbb{N}^{d-1},\ |\beta| \le 2n. \tag{66}$$

Write, using the fundamental theorem of calculus along the segment $[\hat{x},\ \hat{x} + \rho\mathbf{u}(\theta)]$,

$$\chi_h \circ \psi_{\hat{x}}(\rho,\theta) - \chi_h(\hat{x}) = \rho\, \Xi(\rho,\theta), \qquad \Xi(\rho,\theta) := \int_0^1 D\chi_h(\hat{x} + t\rho\mathbf{u}(\theta))\, \mathbf{u}(\theta)\, \mathrm{d}t. \tag{67}$$

Unlike $(\chi_h \circ \psi_{\hat{x}}(\rho,\theta) - \chi_h(\hat{x}))/\rho$, the function $\Xi(\rho,\theta)$ is defined and smooth for every $\rho \ge 0$, *including* $\rho = 0$, where $\Xi(0,\theta) = D\chi_h(\hat{x})\mathbf{u}(\theta)$: (67) is an *exact* identity, not an asymptotic one. Differentiating j times under the integral sign and using (44),

$$\left\|\partial_\rho^j \Xi(\rho,\theta)\right\| = \left\| \int_0^1 t^j D^{j+1}\chi_h(\hat{x} + t\rho\mathbf{u}(\theta))[\mathbf{u}(\theta)]^{\otimes(j+1)}\, \mathrm{d}t \right\| \le \frac{C_Q h^{j+1}}{j+1}, \tag{68}$$

while (61), applied to $\hat{y} = \hat{x} + \rho\mathbf{u}(\theta)$ and letting $\rho \to 0$, gives $\|\Xi(\rho,\theta)\| \ge c_Q h$ for every $\rho \ge 0$. Setting

$$\xi(\rho,\theta) := \frac{\Xi(\rho,\theta)}{h}, \qquad \vartheta(\rho,\theta) := \|\xi(\rho,\theta)\|, \qquad \omega(\rho,\theta) := \frac{\xi(\rho,\theta)}{\vartheta(\rho,\theta)}, \tag{69}$$

we thus have $c_Q \leq \vartheta(\rho,\theta) \leq C_Q$ *uniformly in* h, and, by ordinary (real) Leibniz and Faà di Bruno applied to the norm and quotient maps – legitimate everywhere on $[0,\hat{\rho}] \times [0, 2\pi[$ since ϑ never vanishes – both ϑ and ω, together with all their ρ-derivatives up to order $2n$, are bounded by constants depending only on the derivative order, C_Q and c_Q, again uniformly in h.

Since $\|x-y\| = \rho\|\Xi(\rho,\theta)\| = \rho\, h\, \vartheta(\rho,\theta)$ exactly, substituting (65) into the definition of H_h gives the exact identity

$$H_h(\rho,\theta) = \rho^s \frac{\kappa(x,\omega(\rho,\theta))}{\rho^s h^s \vartheta(\rho,\theta)^s}\,[\hat{u}\circ\psi_{\hat{x}}](\rho,\theta) = h^{-s}\,\mathcal{H}(\rho,\theta), \qquad \mathcal{H}(\rho,\theta) := \frac{\kappa(x,\omega(\rho,\theta))\,[\hat{u}\circ\psi_{\hat{x}}](\rho,\theta)}{\vartheta(\rho,\theta)^s}, \tag{70}$$

in which ρ^s has cancelled *algebraically*, with no asymptotics involved – exactly as at $k=0$, but now at the level of the function itself, before any differentiation. Every factor of $\mathcal{H}$ has ρ-derivatives bounded independently of h: $\omega(\rho,\theta)$ and its derivatives are h-uniformly bounded (above); $\kappa(x,\cdot)$ and its angular derivatives are bounded by (66), so $\rho \mapsto \kappa(x,\omega(\rho,\theta))$ has h-uniformly bounded derivatives by ordinary Faà di Bruno; $[\hat{u}\circ\psi_{\hat{x}}](\rho,\theta)$ has ρ-derivatives bounded by δ_m, exactly as already used above (Lemma 36); and $t \mapsto t^{-s}$ is smooth with bounded derivatives on $[c_Q, C_Q]$, so $\vartheta(\rho,\theta)^{-s}$ and its ρ-derivatives are h-uniformly bounded too. An ordinary (real) application of the Leibniz rule to the product in (70) then yields, for every $k \leq 2n$,

$$\boxed{\gamma_{h,k} \;\leq\; \Lambda_k\, h^{-s}}, \qquad \Lambda_k := \sup_{m\leq k}\sup_{\rho,\theta}\left|\partial_\rho^m \mathcal{H}(\rho,\theta)\right| < \infty, \tag{71}$$

where Λ_k is an explicit, finite, purely combinatorial constant (depending on k, s, C_Q, c_Q, $C_k(\cdot)$ and $\delta_{\leq k}$, but not on h), obtained without ever leaving the reals: no analytic continuation, no complex disk, no Cauchy inequality was used. Setting $k=0$ recovers (64) with $\Lambda_0 = \Gamma_0$, so this construction generalizes the direct computation above to every order at once. (If one further assumes, as is standard for the fundamental solutions used in practice, that $\kappa(x,\cdot)$ and χ_h are real-analytic, Λ_k can in addition be shown to grow no faster than $\Lambda\, k!\; L^k$ for constants Λ, L independent of h and k; this refinement is most conveniently obtained by applying Cauchy's inequality to $\mathcal{H}$ itself – whose domain of analyticity is now transparent from (70) – rather than to H_h directly, but it is not needed for what follows.)

Consequence for the mesh-size convergence rate. Substituting (71) for $\gamma_{h,k+2}$ inside (50), the factor h^{-s} is common to every term of the sum and factors out unconditionally (regardless of the precise growth of Λ_{k+2} with k), so that the h^{d-1} prefactor of (50) is replaced by h^{d-1-s}:

$$|R_h^{(2n)}(\rho)| \leq\; h^{d-1-s}\sum_{k=2n}^{+\infty} k!\;\frac{(k+3)^4 M_{k+2}^2\,\Lambda_{k+2}\,\lambda_{k+2}}{(k+2)!\,(k-2n)!}\,\hat{\rho}^{k-2n}. \tag{72}$$

Since $s \geq 1$ for any genuinely singular kernel, this is strictly worse than the $\mathcal{O}(h^{d-1})$ that the original ansatz $\gamma_{h,k} = \mathcal{O}_{h\to 0}(1)$ would have suggested, and it is *negative* (i.e. $R_h^{(2n)}(\rho_0)$ actually diverges as $h \to 0$ at fixed quadrature order n) as soon as $s > d-1$ – which is precisely the hypersingular case, e.g. $s=3$, $d-1=2$ (see §4.5.1 again: $s=3$, $d=3$). This is not a contradiction: it simply confirms, quantitatively, that mesh refinement at *fixed* n cannot control the error made on a self-singular (diagonal) hypersingular integral – the exponential-in-n decay of the Bernstein ellipse bound of §4.5.1 is what must be relied upon instead, with n allowed to grow (e.g. $n = n(h) \to \infty$, typically $n \sim \log(1/h)$) as the mesh is refined, so that the exponential gain in n outpaces the polynomial-in-$1/h$ growth exhibited here.